



FIRST SEMESTER 2026-27
COURSE HANDOUT (PART II)

Date: 29.07.2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No: ME F323

Course Title: Energy Storage Technologies

Instructor In-Charge: Dr. Srinibas Tripathy

1. Course Description:

This course introduces the necessity, classification, operating principles, and applications of various energy storage technologies used in modern energy systems. Mechanical, thermal, electrochemical, electrical, and chemical energy storage systems will be discussed with emphasis on their design, operation, performance characteristics, advantages, limitations, and practical applications in electric vehicles, renewable energy integration. The course also covers emerging storage technologies including advanced battery systems, supercapacitors, hydrogen energy systems, and phase change material-based thermal storage systems.

2. Scope and Objective of the Course:

The objective of this course is to provide students with a strong understanding of energy storage technologies and their role in sustainable energy systems. The course covers fundamental thermodynamic, electrochemical, and mechanical aspects of energy storage systems along with their design considerations, performance evaluation, safety issues, and practical applications. The course also aims to familiarize students with recent advancements, technical challenges, economic considerations, and future prospects of modern energy storage technologies. The course also introduces Computational Fluid Dynamics (CFD) as a simulation tool for the design, analysis, and performance evaluation of energy storage systems, with emphasis on thermal management and flow behavior.

3. Text Books (T.B.):

- **T.B. 1:** Robert A. Huggins, Energy Storage: Fundamentals, Materials and Applications, Springer, 2nd Edition, 2016.
- **T.B. 2:** Reddy, Linden's Handbook of Batteries, McGraw-Hill Education, 3rd Edition, 2011.

4. Reference Books (R.B.):

- **R.B. 1:** Luisa F. Cabeza, Advances in Thermal Energy Storage Systems: Methods and Applications, Elsevier Science & Technology, ProQuest Ebook Central, 2014.
- **R.B. 2:** Godula-Jopek, Agata, Walter Jehle, and Joerg Wellnitz, Hydrogen Storage Technologies: new materials, transport, and infrastructure, John Wiley & Sons, 2012.
- **R.B. 3:** Sandeep A. Arote. Electrochemical Energy Storage Devices and Supercapacitors: An Overview, Institute of Physics Publishing, ProQuest Ebook Central, 2022.



5. Course Plan:

Learning Objectives	No of Lecture Hour	Reference Chap./Sec. (Book)
Introduction and Overview: Need for energy storage system, classification, principles, challenges, comparison and applications of energy storage technologies, General thermodynamic concepts.	3	T.B. 1: Chapters 1-2 Class Notes
Mechanical Energy Storage System: Flywheel, Gravity, Compressed air, Pumped hydro, design principles, operating characteristics, advantages, limitations and applications.	3	T.B. 1: Chapter 6
Thermal Energy Storage Systems: Sensible heat storage systems, latent heat storage systems, phase change material based thermal storage, thermal performance evaluation and practical applications.	6	T.B. 1: Chapter 3 R.B. 1: Chapters 1-4, 8-13, and 17-23
Thermochemical Energy Storage Systems: Sorption and desorption reactions, adsorption and absorption-based storage systems, thermochemical storage reactions, performance analysis and applications.	2	R.B. 1: Chapters 14 and 15
Electrochemical Energy Storage: Principles, battery terminology, electrochemical reactions, charge-discharge mechanisms, performance parameters and efficiency evaluation.	6	T.B. 2: Chapters 1-3
Battery Technologies: Lead-acid batteries, lithium-ion batteries, sodium-ion batteries, battery management systems, degradation mechanisms and recycling aspects.	6	T.B. 2: Chapters 23 and 35 Class Notes
Chemical Energy Storage and Fuel Cell: Hydrogen production methods, compressed hydrogen storage, liquid hydrogen storage, solid-state hydrogen storage, hydrogen fuel cell fundamentals and hybrid storage systems.	6	T.B. 1: Chapter 8 R.B. 2: Chapters 4-6
Supercapacitors: Battery Vs supercapacitors, electrochemical double layer capacitors and applications.	2	R.B. 2: Chapters 1-3 Class Notes
Emerging Technologies and CFD Modeling: Hybrid energy storage systems, sustainability issues, techno-economic analysis, recycling technologies, circular economy, future scope and industrial challenges, and CFD Modeling of energy storage system.	6	Class Notes and Self-study



6. Evaluation Scheme:

Component	Duration (Minutes)	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90	30	As announced in the timetable	Close Book
Projects, Assignment and Seminars	-	15	TBA in the class	Open or Close Book
Quiz	30	15	TBA in the class	Open or Close Book
Comprehensive Examination	180	40	As announced in the timetable	Open and Close Book

7. Chamber Consultation Hour: 5:00-6:00 PM [Monday]

8. Notices: Notices will be displayed in Nalanda. The students should also check their mails regularly.

9. Make-up Policy: Make-up will be given only to the medical and other emergency cases (check out AUGSD guidelines for more details) for Mid Semester and Comprehensive exams only. The request application for make-up test should reach the instructor-in-charge one day before the commencement of scheduled test (documentary proof is essential).

10. Open Book (OB) Policy: Evaluative components are OB which implies text book only and without any marking and hand-written matter in it and no other books or class notes are allowed.

11. Passing Criteria: Scoring less than 10% of course top score or completely missing any evaluation components of the course e.g., Projects, Assignment, Seminars, Mid-semester test or comprehensive examination will attract NC grade.

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